

Chapter 4: Elementary probability

Definition:

- Probability is the chance of occurring of an outcome of an experiment
- It is the measure of how likely an outcome of an experiment is to occur

Definition of some probability terms

- **Experiment (E)** :is any statistical process that can be repeated several times and in any trial of which the outcome is unpredictable
- It is also called random experiment or probability experiment

Example: If a fair die is rolled once it is possible to list all the possible outcomes

i.e.1, 2, 3, 4, 5, 6 but it is not possible to predict which outcome will occur

...cont

Example of experiment and the possible outcome

- ◁ Tossing a coin only once, $S = \{\text{Head (H), Tail (T)}\}$
- ◁ Tossing a coin two times, $S = \{HH, HT, TH, TT\}$
- ◁ Rolling a die, $S = \{1, 2, 3, 4, 5, 6\}$
- ◁ Selecting an item from a production lot, $S = \{\text{Defective, Non-defective}\}$
- ◁ Introducing a new product, $S = \{\text{Success, Failure}\}$

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Sample Space (S): is a set consisting all possible outcomes of a given experiment

- If a probability experiment has M possible outcomes and done n times, then the sample space (S) has M^n total elements.

Example: What is the sample space for the following experiment

A. Rolling of a die one time.

B. Toss a coin two times

Answer: A) $S = \{1, 2, 3, 4, 5, 6\} = 6^1$ B) $S = (HH, HT, TH, TT) = 2^2$

- **Event:** It is a subset of sample space or

- It is a statement about one or more outcomes of a random experiment

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- They are denoted by capital letters

Example: Considering the above experiment, let A be the event of odd numbers, B be the event of even numbers, and C be the event of number 8. then list the elements of A, B and C

Answer: $A = \{1, 3, 5\}$ $B = \{2, 4, 6\}$ $C = \{8\}$

- **Note:** If sample space (S) has n members, then there are exactly 2^n subsets or events.

Equally Likely Events: Events which have the same chance of occurring.

- **Complement of an Event:** the complement of an event A means non- occurrence of A.
 - It is denoted by A' contains those points of the sample space which don't belong to A

Example: from the above example A is the complement of B and viscera

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- **Elementary Event:** an event having only a single element or sample point.

Example: Let A is the event of getting a number 2 from rolling a die once. $A = \{2\}$

- **Mutually Exclusive Events:** two or more events are mutually exclusive, if they have no outcome in common (i.e. they cannot occur together simultaneously)

Example: Let A is the event of odd numbers and B is the event of even numbers in rolling of a die once. Then we have

$$A = \{1, 3, 5\} \quad B = \{2, 4, 6\}$$

- Here, A and B are mutually exclusive events since $A \cap B = \emptyset$

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- **Independent Events:** Two events are independent if the occurrence of one does not affect the probability of the other occurring.
- **Dependent Events:** Two events are dependent if the first event affects the outcome or occurrence of the second event in a way the probability is changed.

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Counting Rules

- In order to calculate probabilities, we have to know
 - The number of elements of an event
 - The number of elements of the sample space
- In order to determine the number of outcomes, one can use
 - The addition rule
 - The multiplication rule
 - Permutation rule
 - Combination rule
- To list the outcomes of the sequence of events, we can use **tree diagram**

Addition Rule

- **Addition Rule:** states that if a task can be done (accomplished) by any of the k procedures,
- where i^{th} procedures has n_i alternatives, then the total number of ways of doing the task is

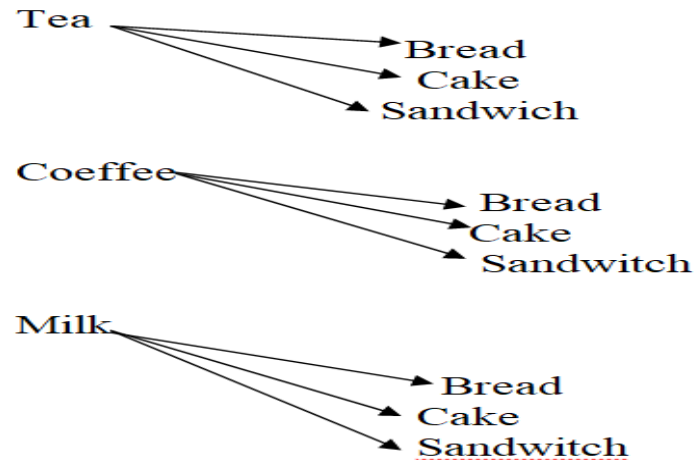
$$n_1 + n_2 + n_3 \dots + n_k = \sum_{i=1}^k n_i$$

Example: A student goes to the nearest Cafe to have a breakfast. He can take tea, coffee, or milk with bread, cake and sand witch. How many possibilities does he have?

Solution: Using tree diagram

The student has

$3 \times 3 = 9$ possibilities



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Multiplication rule

- If a choice consists of k steps of which the first can be made in n_1 ways, the second can be made in n_2 ways..., the k^{th} can be made in n_k ways, then the whole choice can be made in $(n_1 * n_2 * \dots * n_k)$ ways

Example: The digits 0, 1, 2, 3, and 4 are to be used in 4 digit identification card.

How many different cards are possible if

- A. Repetitions are permitted
- B. Repetitions are not permitted

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Answer: A)

1 st digit	2 nd digit	3 rd digit	4 th digit
5	5	5	5

There are four steps

Step 1: Selecting the first digit, this can be made in 5 ways

Step 2: Selecting the second digit, this can be done in 5 ways

Step 3: Selecting the third digit, this can be made in 5 ways

Step 4: Selecting the fourth digit, it can be made in 5 ways

$\Rightarrow 5*5*5*5 = 625$ different cards can be made

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B)

1 st digit	2 nd digit	3 rd digit	4 th digit
5	4	3	2

Step 1: Selecting the first digit, this can be made in 5 ways

Step 2: Selecting the second digit, this can be done in 4 ways

Step 3: Selecting the third digit, this can be made in 3 ways

Step 4: Selecting the fourth digit, it can be made in 2 ways

$\Rightarrow 5*4*3*2 = 120$ different cards can be made

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Permutation

- An arrangement of n objects in a specified order is called permutation of the objects

Permutation Rules:

1. The number of permutations of n distinct objects taken all together is $n!$

Where $n! = n * (n - 1)! * (n - 2)! * \dots * 3 * 2 * 1$

2. The arrangement of n objects in a specified order using r objects at a time is called the permutation of n objects taken r objects at a time.

- It is written as $P(n, r) = \frac{n!}{(n-r)!}$

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3. The number of permutations of n objects in which k_1 are alike k_2 are alike ---- etc is

$$P(n, r) = \frac{n!}{k_1! * k_2! * \dots * k_n!}$$

Examples:

1. Suppose we have a letters A, B, C, D

A. How many permutations are there taking all the four?

B. How many permutations are there two letters at a time?

2. How many different permutations can be made from the letters in the word "CORRECTION"

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Answer: 1)

A) $n = 4 \Rightarrow 4! = 4*3*2*1 = 24$ different permutations

B) $n = 4, r = 2$

$$\Rightarrow P(n, r) = P(4, 2) = \frac{4!}{(4-2)!} = \frac{4*3!}{2!} = 4 \text{ different permutations}$$

2) Here $n = 10$, 2 are C, 2 are O, 2 are R, 1E, 1T, 1I 1N

$$\Rightarrow K_1 = 2, K_2 = 2, K_3 = 2, K_4 = 1, K_5 = 1, K_6 = 1, K_7 = 1$$

$$\begin{aligned} \Rightarrow \frac{10!}{2!*2!*2!*1!*1!*1!*1!} &= \frac{10*9*8*7*6*5*4*3*2!}{2!*2!*2!*1!*1!*1!*1!} = 10*9*7*6*5*4*3*2 \\ &= 453600 \text{ permutations} \end{aligned}$$

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Combination

- A selection of objects with out regard to order is called combination.

Example: Given the letters A, B, C, and D list the permutation and combination for selecting two letters.

Answer:

Permutation

AB BA CA DA

AC BC CB DB

AD BD CD DC

Combination

AB BC AC

BD AD DC

Note: In permutation AB is different from BA. But in combination AB is the same as BA.

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Combination rule

Rule 1: The number of combinations of r objects selected from n objects is denoted by

$$C\binom{n}{r} = \frac{n!}{(n-r)!r!}$$

Example: In how many ways a committee of 5 people be chosen out of 9 people?

Answer: $n = 9, r = 5$

$$C\binom{n}{r} = C\binom{9}{5} = \frac{9!}{(9-5)!5!} = \frac{9!}{4!5!} = \frac{9*8*7*6*5!}{4!*5!} = \frac{9*8*7*6}{4*3*2*1} = 9*2*7 = 126 \text{ ways}$$

- **Rule 2:** If the selection has k steps, by selecting r_1 of n_1 objects, r_2 of n_2 , ..., r_k of n_k objects, then the total number of ways of doing this selection is equal to

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$$\binom{n_1}{r_1} \times \binom{n_2}{r_2} \times \dots \times \binom{n_k}{r_k}$$

Example: Among 15 clocks there are two defectives .In how many ways can an inspector chose three of the clocks for inspection so that:

- A. There is no restriction
- B. None of the defective clock is included
- C. Only one of the defective clocks is included
- D. Two of the defective clock is included

Answer

n = 15 clocks, defective clock =2, non-defective clocks = 13

A. If there is no restriction select three clocks from 15 clocks and this can be done in

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$$n = 15, r = 3$$

$$\binom{n}{r} = \frac{n!}{(n-r)!r!} \Rightarrow \binom{15}{3} = \frac{15!}{(15-3)!3!} = \frac{15*14*13*12!}{12!3!} = \frac{15*14*13}{6} = 2730/6 = 455$$

B. None of the defective clocks is included

- This is equivalent to zero defective and three non defective, which can be done in

$$C\binom{2}{0} * C\binom{13}{3} = \frac{2!}{(2-0)!0!} * \frac{13!}{(13-3)!3!} = \frac{2!}{2!0!} * \frac{13!}{10!3!}$$

$$= \frac{2*1}{2*1} * \frac{13*12*11*10!}{10!3*2} = 1 * \frac{1716}{6} = 286 \text{ possible ways}$$

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C. Only one of the defective clocks is included.

- This is equivalent to one defective and two non defective, which can be done in:

$$C\left(\begin{matrix} 2 \\ 1 \end{matrix}\right) * C\left(\begin{matrix} 13 \\ 2 \end{matrix}\right) = \frac{2!}{(2-1!)1!} * \frac{13!}{(13-2)!2!} = \frac{2!}{1!1!} * \frac{13!}{11!2!}$$

$$= 2 * \frac{13*12*11!}{11!2!}$$

$$= 2 * \frac{156}{2}$$

$$= 156 \text{ possible ways}$$

Con...

D. Two of the defective clock is included.

- This is equivalent to two defective and one non defective, which can be done in:

$$C\binom{2}{2} * C\binom{13}{1} = \frac{2!}{(2-2!)2!} * \frac{13!}{(13-1)!1!} = \frac{2!}{0!2!} * \frac{13!}{12!1!}$$

$$= 1 * \frac{13*12!}{12!1!}$$

$$= 1*13$$

$$= 13 \text{ possible ways}$$

Some Probability Rules/Axioms

- Let S be a sample space associated with a random experiment. Then with any event E , in this sample space, we associate a real number called probability of E satisfying the following properties (axioms).

- ▶ $0 \leq P(E) \leq 1$

- ▶ $P(S) = 1$

- ▶ If A and B are mutually exclusive events, then

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$$

- ▶ If $A_1, A_2, \dots, A_n$ are pairwise mutually exclusive events, then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

- ▶ $P(A \cup A^c) = P(A) + P(A^c)$

...cont

► $P(\phi) = 0$

Using the above axioms, it can be shown that for any two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example 1: A box of 20 candles consists of 5 defective and 15 non-defective candles. If 4 of these candles are selected at random, what is the probability that

(a) all will be defective.

Solution:

Let A be an event of all candles are defective.

$$P(A) = \frac{n(A)}{n(S)} = \frac{\binom{5}{4} \times \binom{15}{0}}{\binom{20}{4}} = 0.001032$$

(b) 3 will be non-defective.

Let B be an event of 3 candles are non-defective.

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Let B be an event of 3 candles are non-defective.

$$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{5}{1} \times \binom{15}{3}}{\binom{20}{4}} = 0.4696$$

(c) all will be non-defective.

Let C be an event of all candles are non-defective.

$$P(C) = \frac{n(C)}{n(S)} = \frac{\binom{5}{0} \times \binom{15}{4}}{\binom{20}{4}} = 0.2817$$

Example 2: An urn contains 6 white, 4 red and 9 black balls. If 3 balls are drawn at random, find the probability that

(a) two of the balls drawn are whites.

Let E_1 be an event two of the balls drawn are whites.

$$P(E_1) = \frac{n(E_1)}{n(S)} = \frac{\binom{6}{2} \times \binom{13}{1}}{\binom{19}{3}} = 0.2012$$

...cont

(b) one is from each colour.

Let E_2 be an event of one from each colour.

$$P(E_2) = \frac{n(E_2)}{n(S)} = \frac{\binom{6}{1} \times \binom{4}{1} \times \binom{9}{1}}{\binom{19}{3}} = 0.2229$$

(c) none is red.

Let E_3 be an event of none is red.

$$P(E_3) = \frac{n(E_3)}{n(S)} = \frac{\binom{15}{3} \times \binom{4}{0}}{\binom{19}{3}} = 0.4695$$

(d) at least one is white.

Let E_4 be an event of at least one is white.

$$P(E_4) = \frac{n(E_4)}{n(S)} = \frac{\binom{6}{1} \times \binom{13}{2}}{\binom{19}{3}} + \frac{\binom{6}{2} \times \binom{13}{1}}{\binom{19}{3}} + \frac{\binom{6}{3} \times \binom{13}{0}}{\binom{19}{3}} = 0.7048$$

Random variables and probability distribution

❑ What is random variable?

- A **random** variable is a numerical description of the outcomes of the experiment.
 - or it is a numerical valued function defined on sample space, usually denoted by capital letters.
- A random variable takes a possible outcome and assigns a number to it.

Example: Flip a coin three times, let X be the number of heads in three tosses.

A. Find the sample space(S) B. Find the possible values of X

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Soln: A. $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

A. $X(HHH) = 3, X(HHT) = X(HTH) = X(THH) = 2, X(HTT) = X(THT) = X(TTH) = 1, X(TTT) = 0$

Then the possible values of X are

$$\Rightarrow X = (0, 1, 2, 3)$$

□ What is probability distribution?

- **Probability distribution** is a distribution which consists of a value a random variable can assume and the corresponding probabilities of the values

Example: Consider the experiment of tossing a coin three times. Let X be the number of heads. Then construct the probability distribution of X.

solution

1. First identify the possible value that X can assume.
2. Calculate the probability of each possible distinct value of X and express X in the form of frequency distribution

$X = x$	0	1	2	3
$P(X = x)$	$1/8$	$3/8$	$3/8$	$1/8$

- Probability distribution is denoted by P for discrete and by f for continuous random variable.

Properties of Probability Distribution

- $P(x) \geq 0$ if X is discrete random variable
- $f(x) \geq 0$, if X is continuous variable
- $\sum_x P(X = x) = 1$, if X is discrete random variable
- $\int f(x)dx = 1$ if X is continuous random variable
- If X is continuous random variable, then
 - $P(a < X < b) = \int_a^b f(x)dx$

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- $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$

- If X is discrete random variable then

$$P(a < X < b) = \sum_{x=a+1}^{b-1} P(x)$$

$$P(a \leq X < b) = \sum_{x=a}^{b-1} p(x)$$

$$P(a < X \leq b) = \sum_{x=a+1}^b P(x)$$

$$P(a \leq X \leq b) = \sum_{x=a}^b P(x)$$

Con...

- Probability means area for continuous random variable

- that means, $P(a < X < b) = \int_a^b f(x)dx$

Introduction to expectation

- Let a discrete random variable X assume the values $X_1, X_2, \dots, X_n$ with the probabilities $P(X_1), P(X_2), \dots, P(X_n)$ respectively.
- Then the expected value of X , denoted as $E(X)$ is defined

$$E(X) = X_1 P(X_1) + X_2 P(X_2) + \dots + X_n P(X_n) = \sum_{i=1}^n X_i P(X_i)$$

Con...

- Let X be a continuous random variable assuming the values in the interval (a, b) such that

$$\int_a^b f(x)dx = 1, \text{ then}$$

$$E(X) = \int_a^b xf(x)dx$$

Examples: What is the expected value of a random variable X obtained by tossing a coin three times where X is the number of heads

Soln: First construct the probability distribution of X

$X = x$	0	1	2	3
$P(X = x)$	1/8	3/8	3/8	1/8

Con...

$$\Rightarrow E(X) = X_1 P(X_1) + X_2 P(X_2) + X_3 P(X_3) + X_4 P(X_4)$$

$$= 0 \cdot 1/8 + 1 \cdot 3/8 + 2 \cdot 3/8 + 3 \cdot 1/8$$

$$= 3/8 + 6/8 + 3/8$$

$$= 12/8$$

$$= 1.5 \text{ /////}$$

Mean and Variance of a random variable

- Let X be given random variable, then
- The expected value of X is its mean

$$\Rightarrow \text{Mean of } X = E(X)$$

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- The variance of X is given by

$$\text{Variance of } X = \text{var}(X) = E(X^2) - [E(X)]^2$$

Where, $E(X^2) = \sum_{i=1}^n x_i^2 P(X = x_i)$ if X is discrete random variable

$$E(X^2) = \int x^2 f(x) dx \text{ if } X \text{ is continuous variable}$$

Example 2: Suppose that X is a continuous random variable with pdf of

$$f(x) = \begin{cases} 1 + x, & -1 \leq x < 0 \\ 1 - x, & 0 \leq x \leq 1 \end{cases}$$

then find the mean value and variance of X .

...cont

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_{-1}^0 x(1+x) dx + \int_0^1 x(1-x) dx$$

$$E(X) = \left[\frac{x^2}{2} + \frac{x^3}{3} \right]_{-1}^0 + \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = -\frac{1}{6} + \frac{1}{6} = 0$$

$$E(X^2) = \int_{-1}^0 x^2(1+x) dx + \int_0^1 x^2(1-x) dx$$

$$E(X^2) = \left[\frac{x^3}{3} + \frac{x^4}{4} \right]_{-1}^0 + \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = \frac{1}{12} + \frac{1}{12} = 0.167$$

$$E(X^2) = \left[\frac{x^3}{3} + \frac{x^4}{4} \right]_{-1}^0 + \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = \frac{1}{12} + \frac{1}{12} = 0.167$$

$$\sigma_x^2 = E[X - E(X)]^2 = E[X - \mu]^2 = E(X^2) - \mu^2$$

$$\sigma_x^2 = E(X^2) - \mu^2 = 0.167 - 0^2 = 0.167$$

...cont

Exercise: 1. A coin is tossed two times. Let X be the number of heads. Find the mean value and the standard deviation of X .

2. Suppose that X is a continuous random variable with pdf of

a. $f(x) = 1 + x, -1 \leq x < 3$

b. $f(x) = 2x, 1 \leq x < 3$

Example: What is the variance of a random variable X obtained by tossing a coin three times where X is the number of heads.

Soln: $E(X) = 1.5$ (from the above example)

$$E(X^2) = \sum_{i=1}^n x_i^2 P(X = x_i) = 0^2 * 1/8 + 1^2 * 3/8 + 2^2 * 3/8 + 3^2 * 1/8$$

Con...

$$= 3/8 + 12/8 + 9/8 = 24/8 = 3////$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= 3 - (1.5)^2$$

$$= 3 - 2.25$$

$$= 0.75////$$